Basic Electrical Engineering (EE1100)

Credits Hours: 3+1, Outside Class Hours: 6, Total credits: 10
Class Hours: Slot E, Tue.: 11.00-11.50, Wed.: 10.00-10.50, Thu.: 08.00-08.50, Fri.: 5.00-5.50
Marks Structure: Q-I: 20, Q-II: 20, End Term: 60,
Quiz I: August 27, Quiz II: October 08, End Sem.: November 17

Room Number: CRC 201
Teacher: Prof. Mahesh Kumar

Course Contents

1. Properties of resistance, Ohms law, KVL, KCL, mesh and nodal analysis, Network theorems: Superposition, Thevenin, Norton and maximum power transfer.

2. Properties of inductance and capacitance, DC transients: Series RL, RC, RLC and parallel RLC.

3. Single phase AC, voltage and current phasors, impedance, network theorems application to AC, frequency response of ac circuits, resonance, filters, active power, reactive power, apparent power, power factor.

4. Balanced Three phase AC, three-phase power, star, and delta connection.

5. Single phase transformer: Principle of operation, equivalent circuit, OC and SC test, voltage regulation, efficiency.

6. Three-phase Induction motor: Construction, rotating magnetic field, principle of operation, slip, torque, equivalent circuit, efficiency.

7. DC machines: Principle of operation, excitation, equivalent circuit, emf, speed and torque characteristics.

8. Diodes and applications: Diode characteristics, voltage and current relationship, diode circuits-rectifiers, peak and envelop detectors, solar cell.

9. Operational amplifiers: Description of amplifiers as a black box and definition of gain, effect of feedback on gain, Operational amplifier circuits: Non-inverting, inverting, summing, differential, integrators, differentiators, buffers.

Text Book

Electrical Engineering Fundamentals, Vincent Del Toro, Prentice Hall, 2006.

Reference Books

[1] Electrical Circuit Theory and Technology, John Bird, Elsevier, 2011.

[2] C. K. Alexander and M. N. O. Sadiku, Fundamentals of Electric Circuits, 4th ed. New York, NY, USA: McGraw-Hill Higher Education, 2009.

[3] Essentials of Electrical and Computer Engineering, Kerns & Irwin, Pearson, 2004

[4] Electrical Engineering Concepts and Applications, Carlson and Gisser, Addison Wesley, 1990.